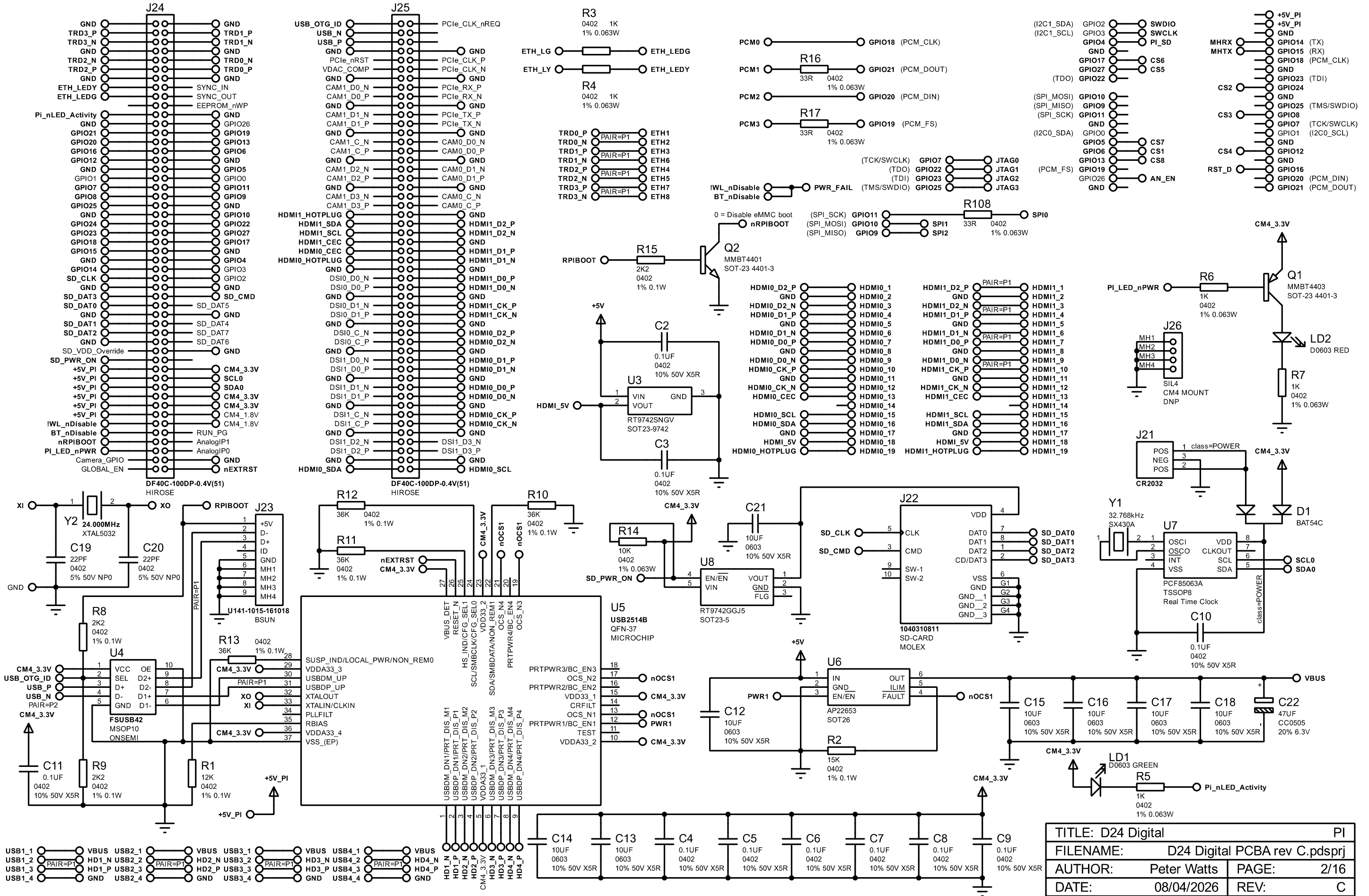
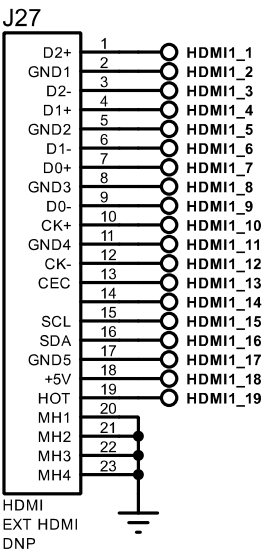
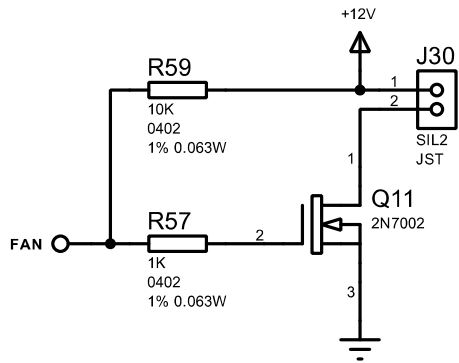
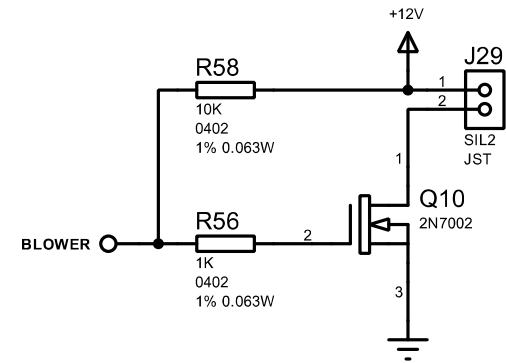
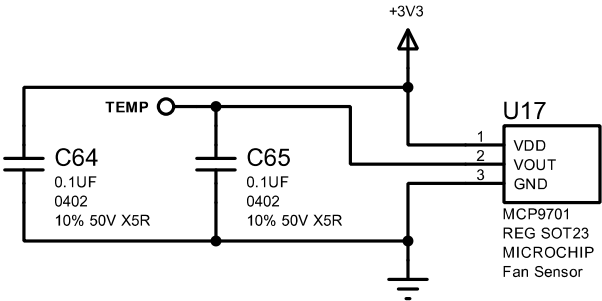




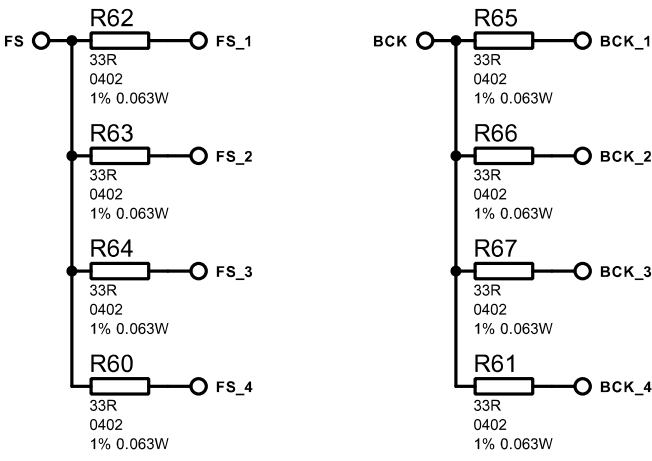
TITLE: D24 Digital		PI
FILENAME: D24 Digital PCBA rev C.pdsprj		
AUTHOR: Peter Watts	PAGE: 2/16	
DATE: 08/04/2026	REV: C	



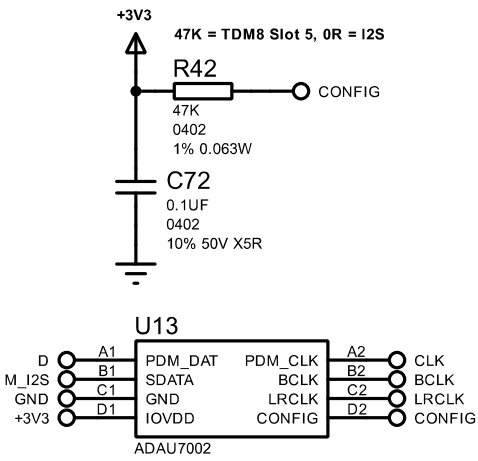
TITLE: D24 Digital		HDMI EXT	
FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 3/16
DATE:		08/04/2026	REV: C



TITLE: D24 Digital		FAN	
FILENAME: D24 Digital PCBA rev C.pdsprj			
AUTHOR: Peter Watts		PAGE: 4/16	
DATE: 08/04/2026		REV: C	

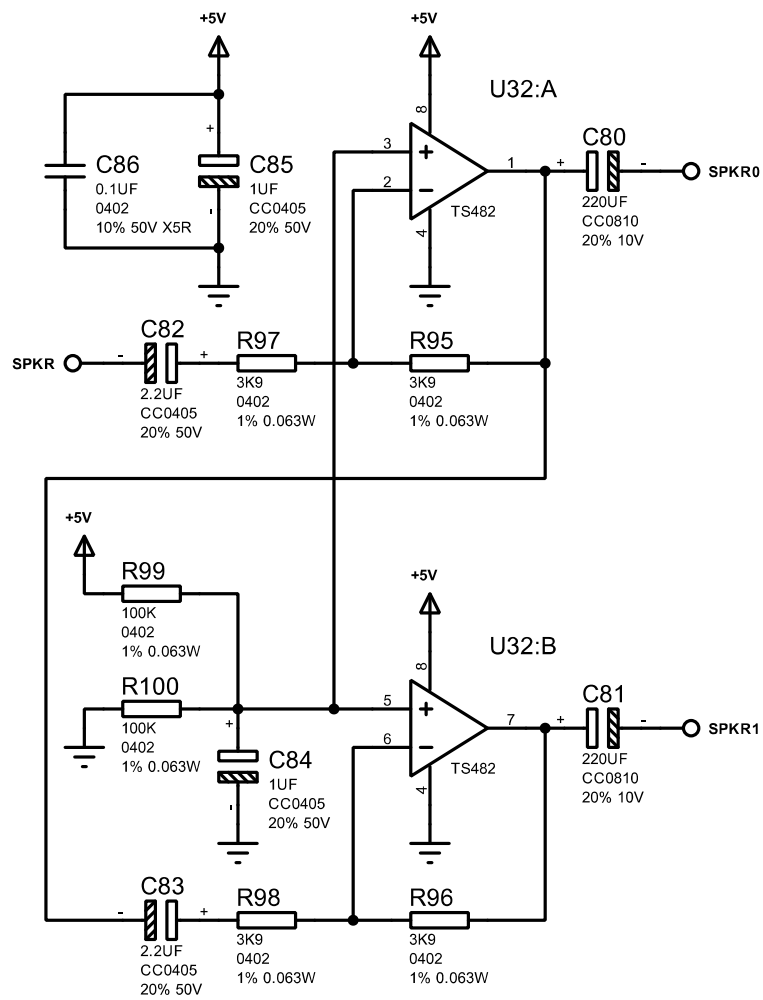
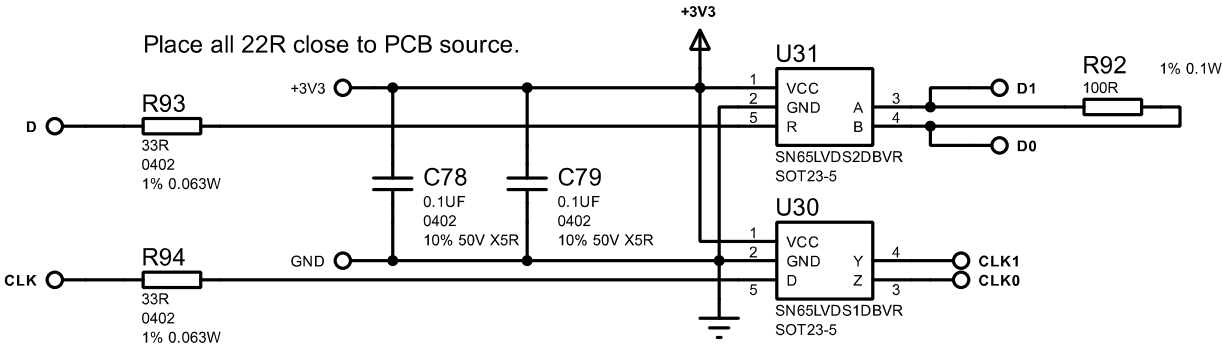


TITLE: D24 Digital		R	
FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 5/16
DATE:		08/04/2026	REV: C

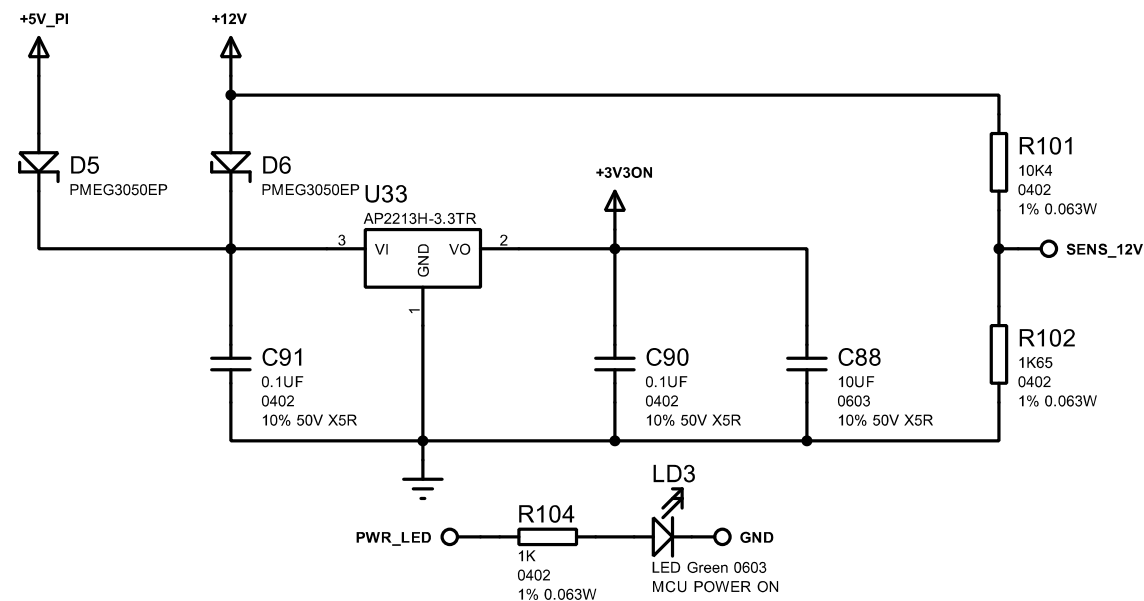


NOTES

Place all 22R close to PCB source.



TITLE: D24 Digital		MEMS	
FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 6/16
DATE:		08/04/2026	REV: C

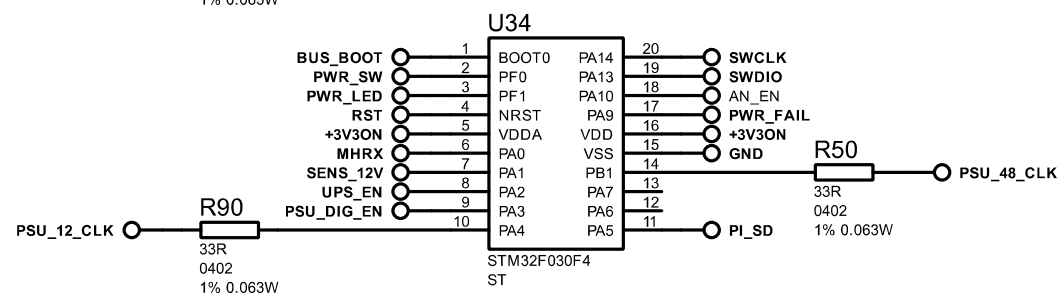
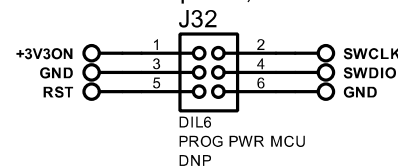


PROG HEADER

- 1 VDD
- 2 SWCLK
- 3 GND
- 4 SWDIO
- 5 RST
- 6 GND



Optional programming using pins 2, 4 & 6.



MCU POWER notes:

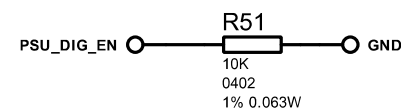
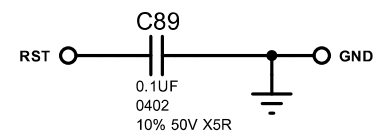
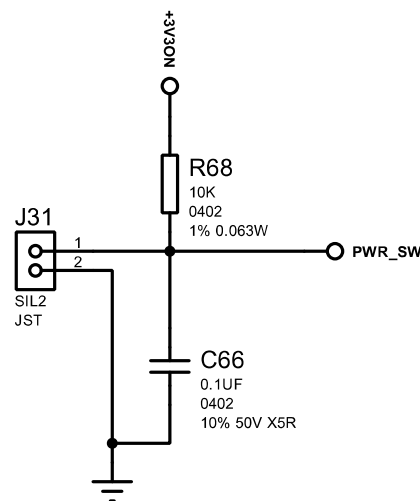
Power Control MCU is always on.

MCU DC CONVERTER ENABLES

- 1. UPS_EN = 5V5 and 5V_Pi enables
- 2. PSU_DIG_EN = All Digital enables
- 3. PSU_12_CLK = 1MHz 20% duty
- 4. PSU_48_CLK = 1MHz 80% duty

SENS_12V checks if 12V PSU is on
PI_SD hi = request to Pi to save and shutdown, via software poll???
AN_EN hi = request from Pi to enable analog power, and disable boot blinking.
PWR_FAIL 0 = disables BT and Wifi, during save and shutdown.
PWR_LED Hi = pwr on, should blink during boot and shutdown.

Pi UART0_TX (MHRX) hi = indicates Pi is powered on.

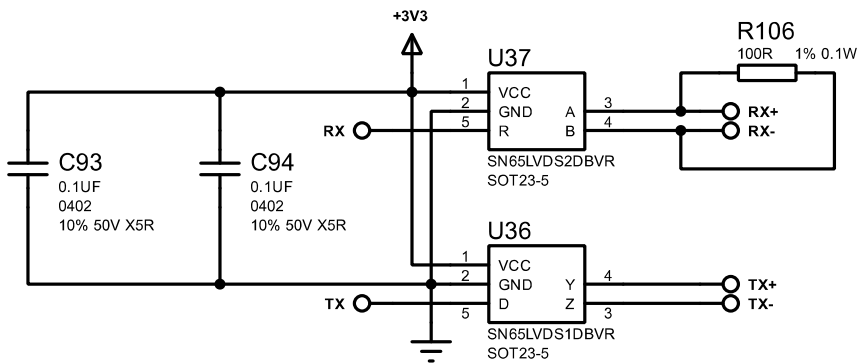
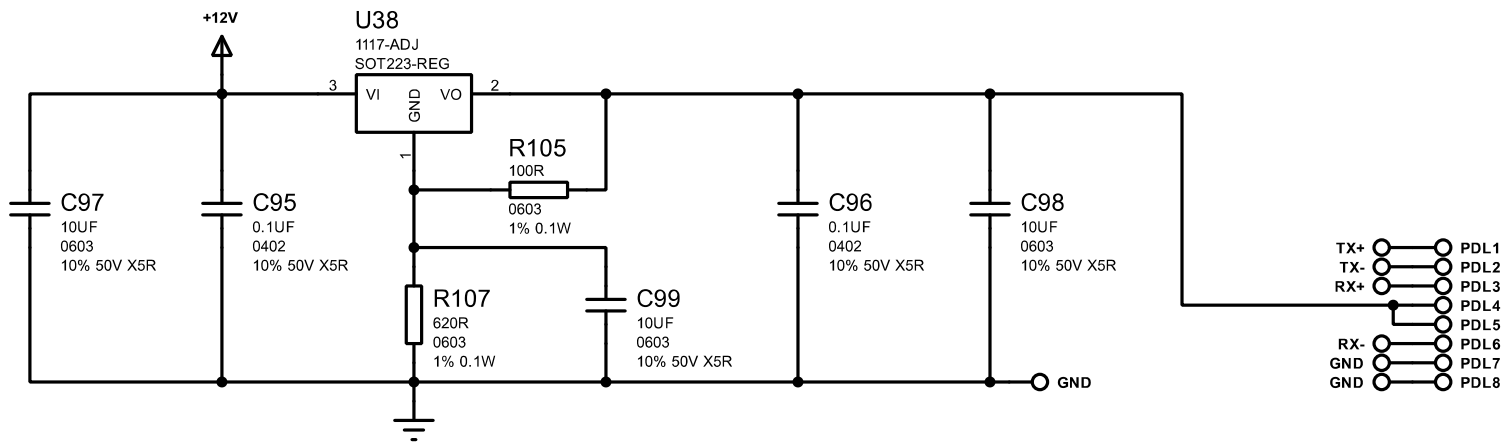


TITLE: D24 Digital		PWR_SW	
FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 7/16
DATE:		08/04/2026	REV: C

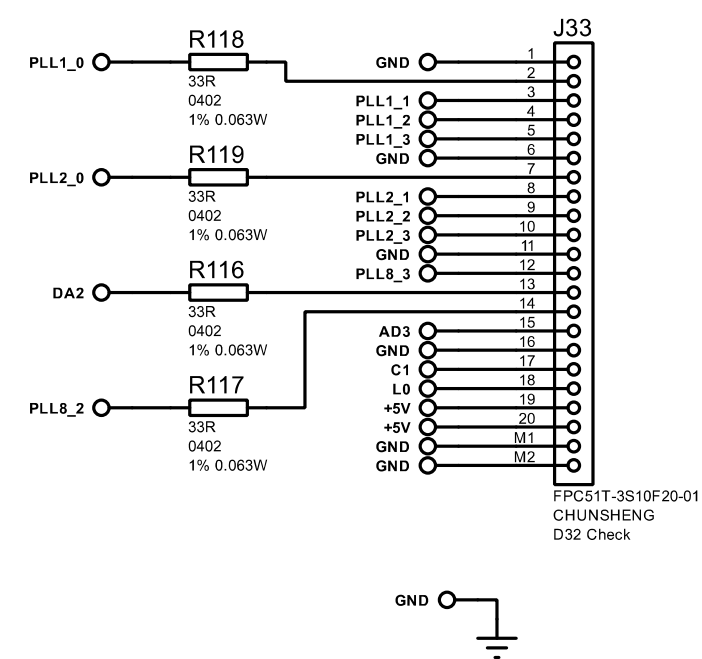
1117 adjustable regulator

Output voltage =
 $1.25 \times (1 + R_{\text{gnd}}/R_{\text{fb}})$

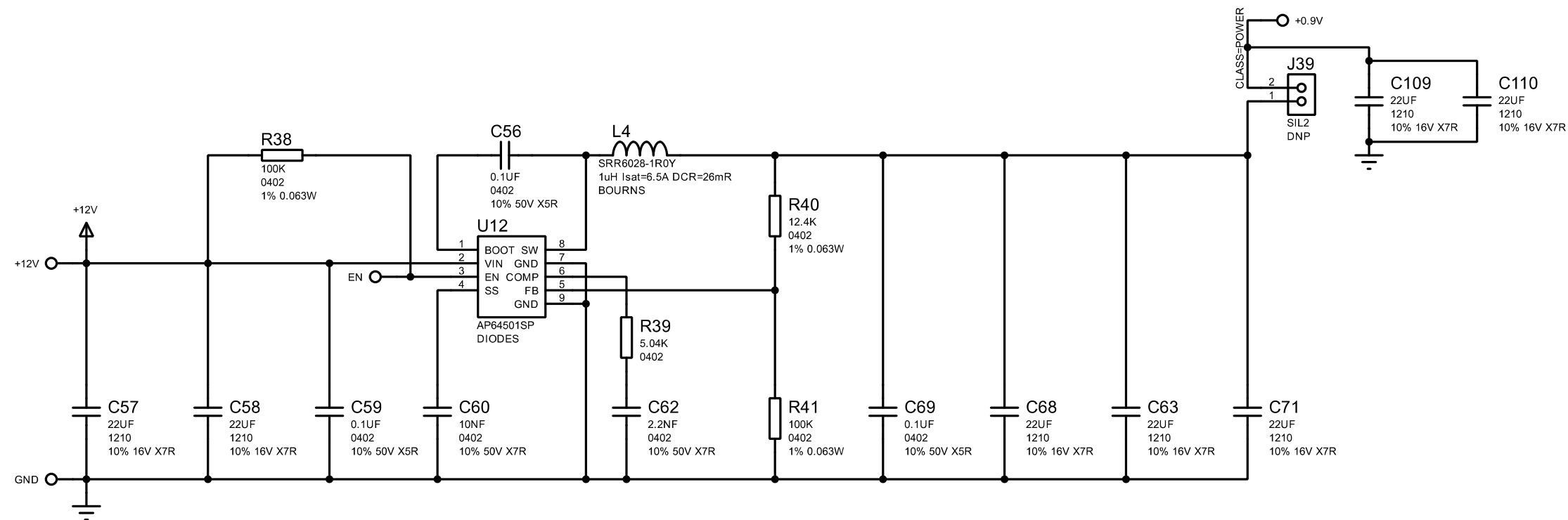
Example:
 $1.25 \times (1 + 620/100) = 9\text{V}.$



TITLE: D24 Digital		PEDAL	
FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 8/16
DATE:		08/04/2026	REV: C



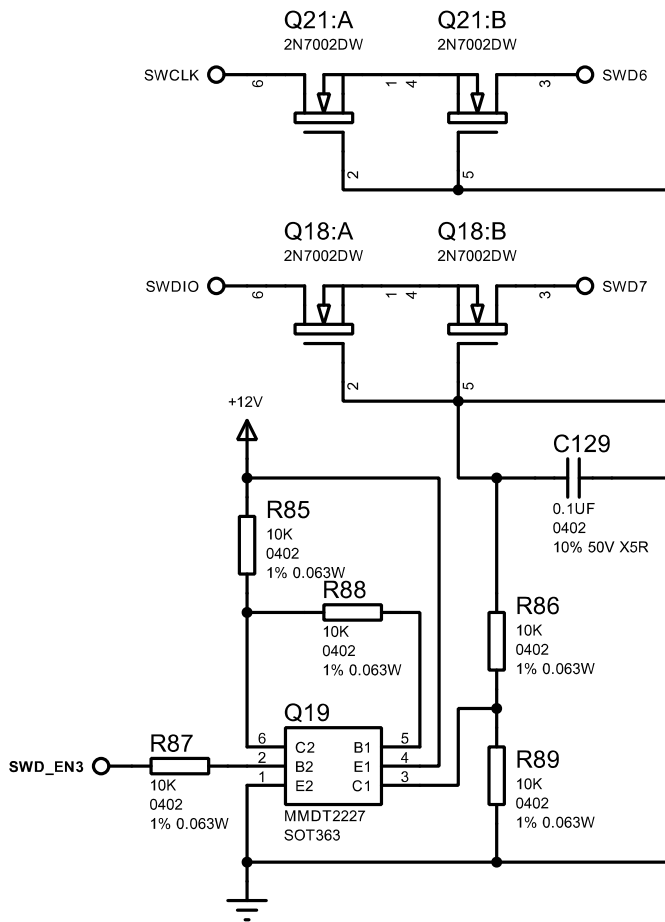
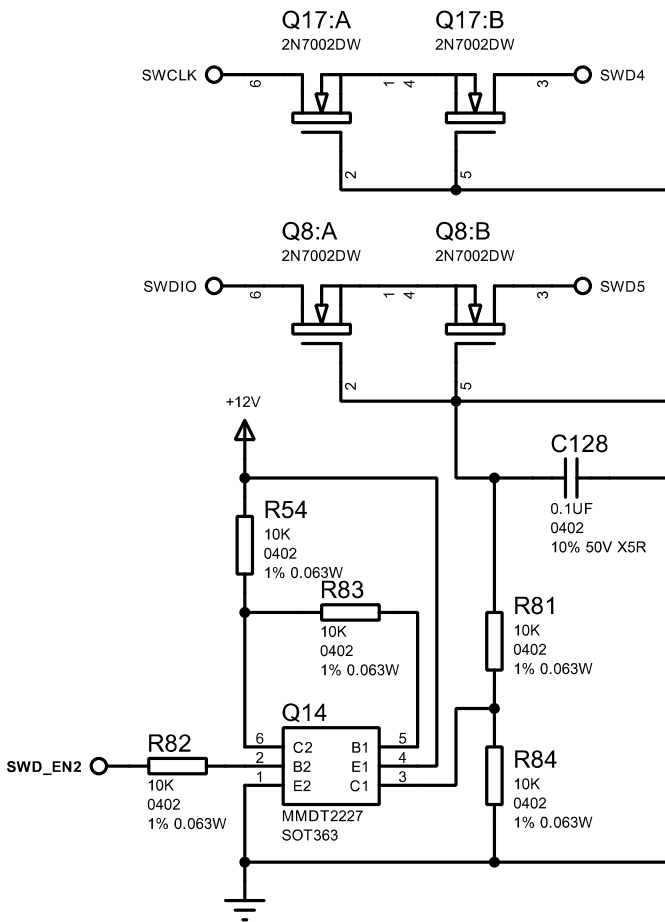
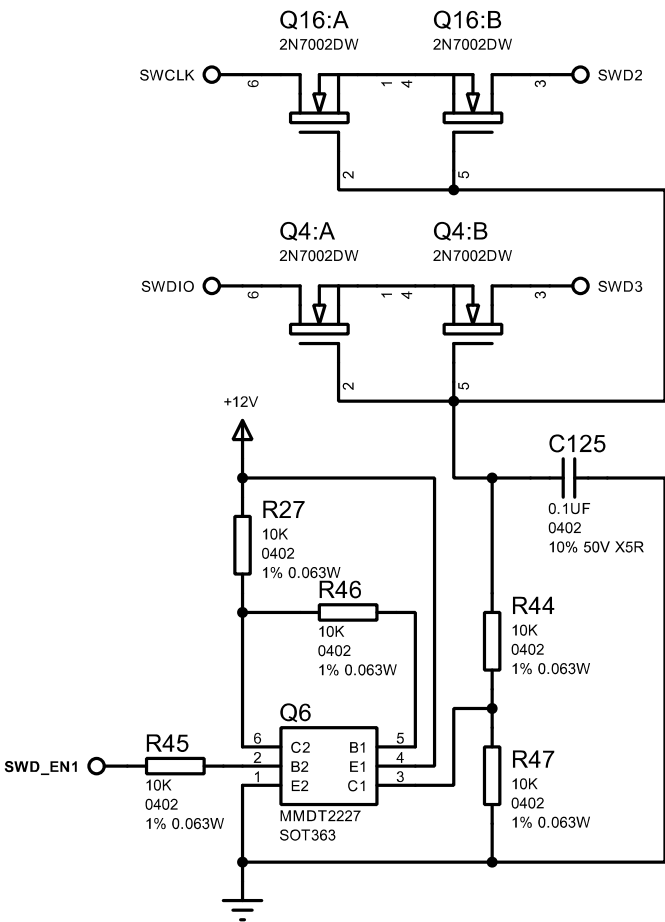
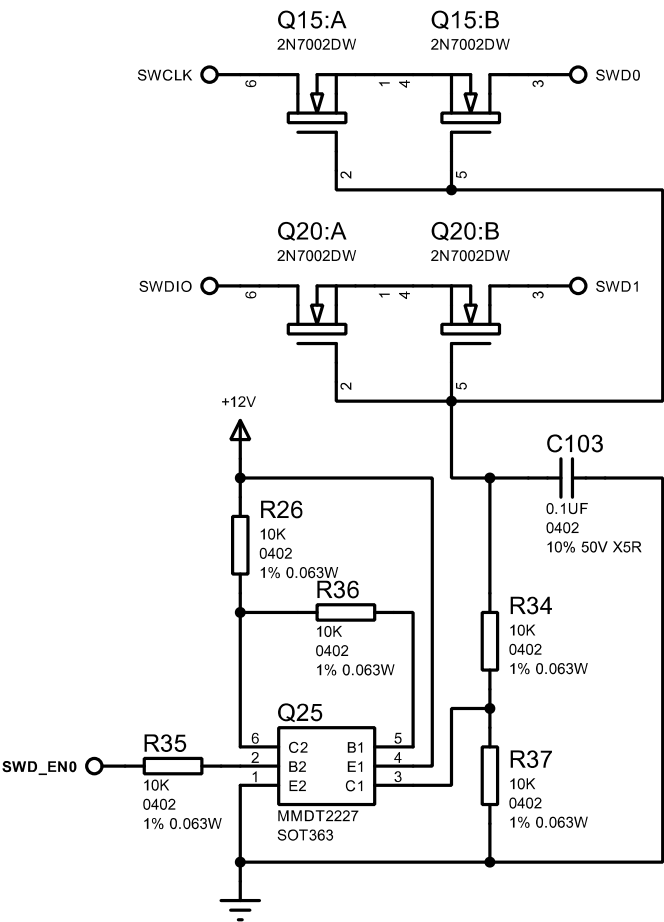
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FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 9/16
DATE:		08/04/2026	REV: C



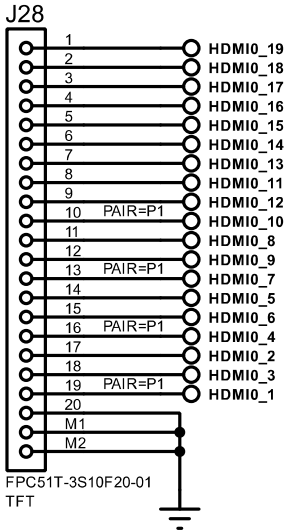
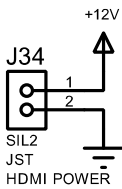
RES VALUES FOR OUTPUT VOLTAGES
INPUT VOLTAGE 3.8V - 40V

OUTPUT	R1	R5	L
1.2V	11K	3.74K	1.5uH
1.5V	19.6K	4.75K	2.2uH
1.8V	27.4K	5.62K	2.2uH
2.5V	47.5K	7.87K	3.3uH
3.3V	69.8K	10.5K	3.3uH
5.0V	115.8K	15.8K	3.6uH
12V	309K	37.4K	10uH
5.5V	130K	17.14K	3.6uH
6V	143K	18.7K	3.6uH

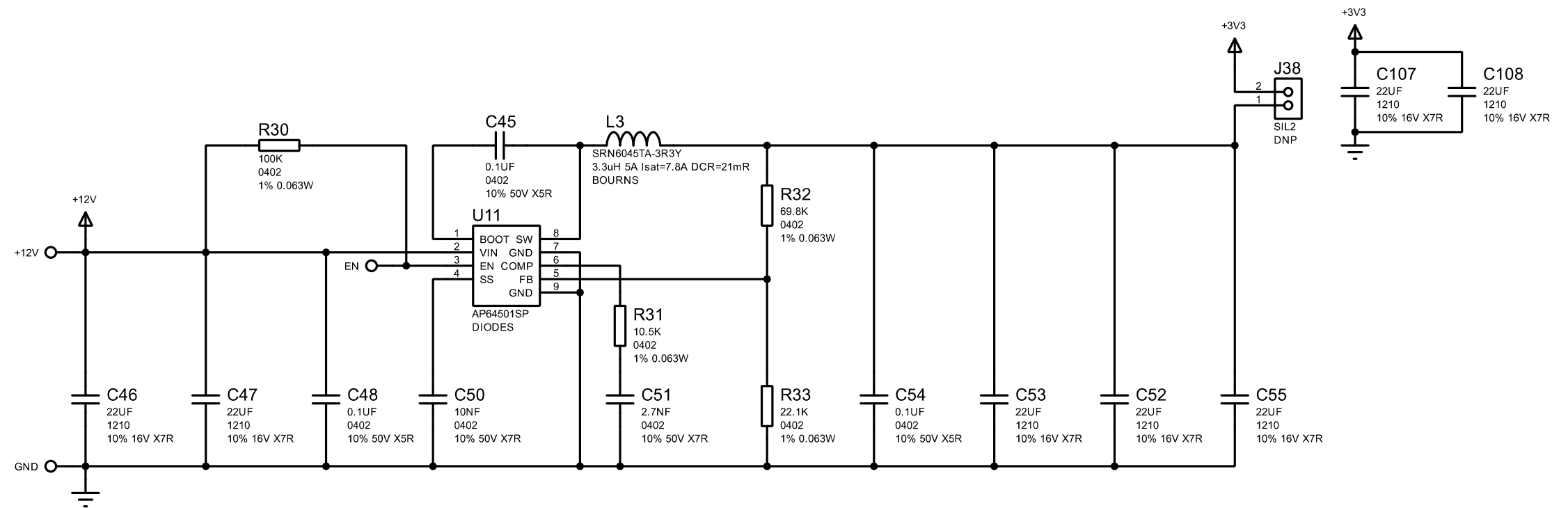
TITLE: D24 Digital		+0.9V	
FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 10/16
DATE:		08/04/2026	REV: C



TITLE: D24 Digital		OPT_SWD	
FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 11/16
DATE:		08/04/2026	REV: C



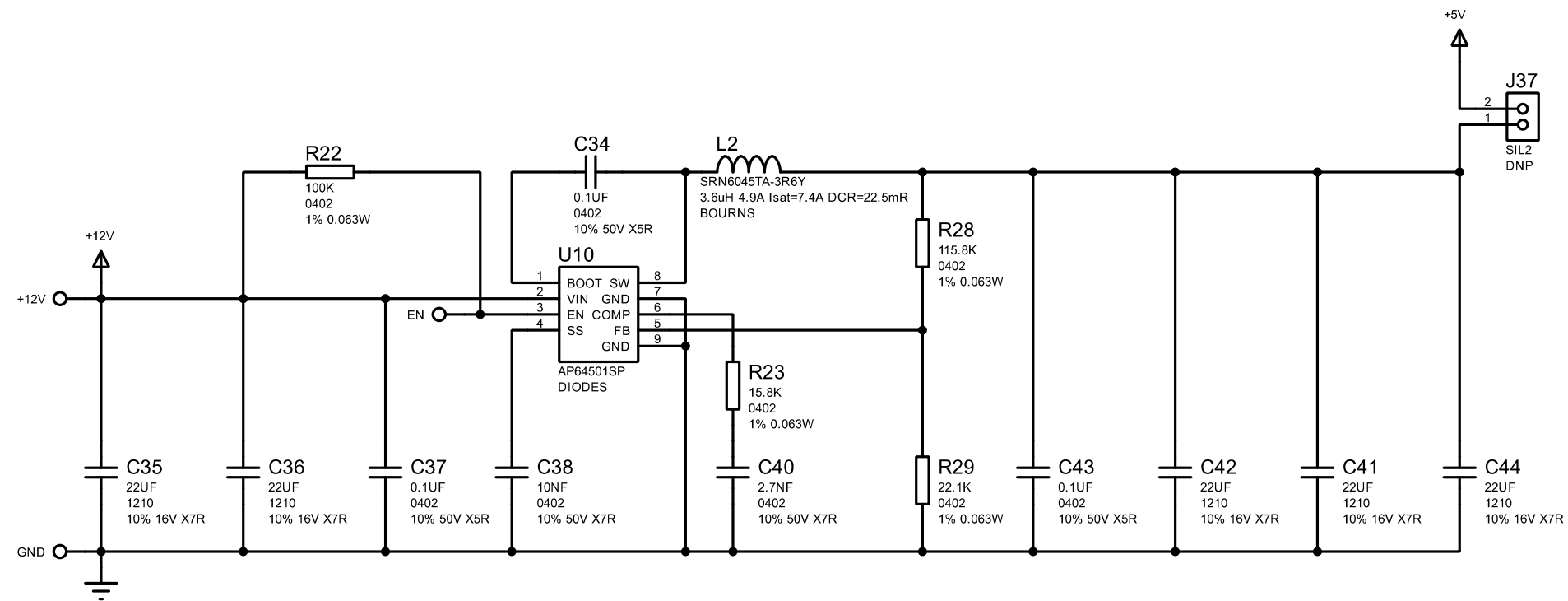
TITLE: D24 Digital		TFT	
FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 12/16
DATE:		08/04/2026	REV: C



RES VALUES FOR OUTPUT VOLTAGES
INPUT VOLTAGE 3.8V - 40V

OUTPUT	R1	R5	L
1.2V	11K	3.74K	1.5uH
1.5V	19.6K	4.75K	2.2uH
1.8V	27.4K	5.62K	2.2uH
2.5V	47.5K	7.87K	3.3uH
3.3V	69.8K	10.5K	3.3uH
5.0V	115.8K	15.8K	3.6uH
12V	309K	37.4K	10uH
5.5V	130K	17.14K	3.6uH
6V	143K	18.7K	3.6uH

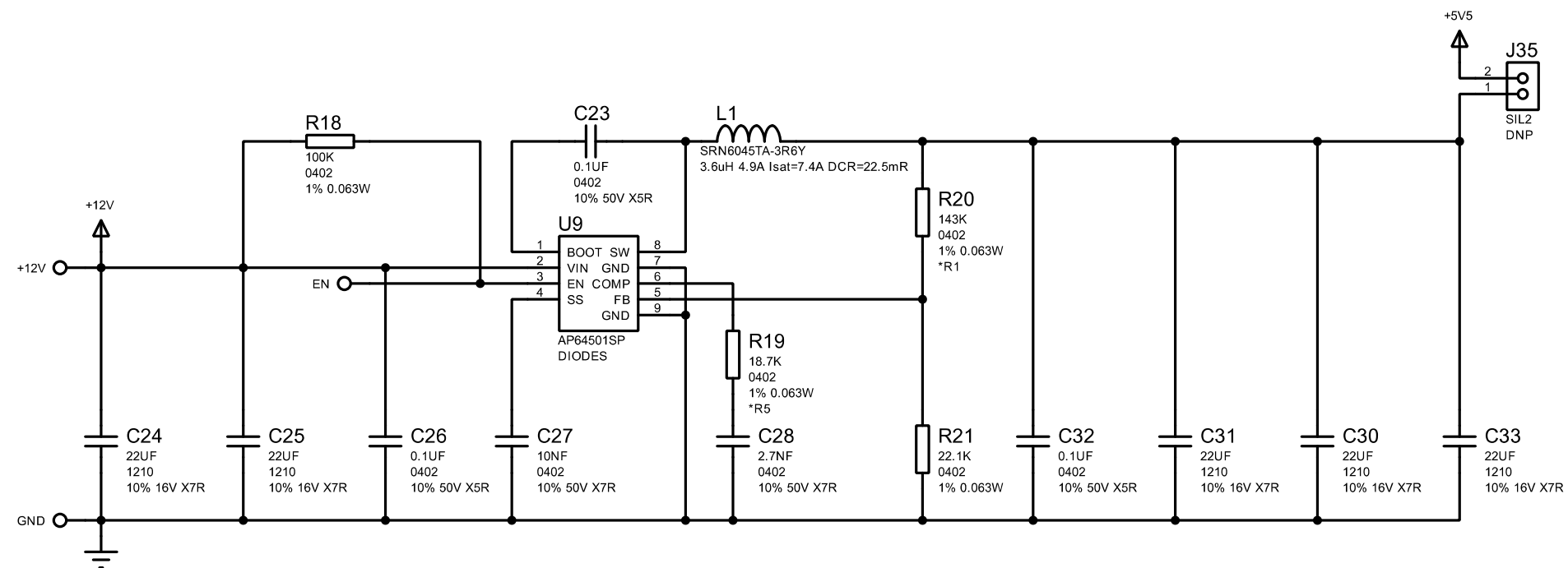
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FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 13/16
DATE:		08/04/2026	REV: C



RES VALUES FOR OUTPUT VOLTAGES
INPUT VOLTAGE 3.8V - 40V

OUTPUT	R1	R5	L
1.2V	11K	3.74K	1.5uH
1.5V	19.6K	4.75K	2.2uH
1.8V	27.4K	5.62K	2.2uH
2.5V	47.5K	7.87K	3.3uH
3.3V	69.8K	10.5K	3.3uH
5.0V	115.8K	15.8K	3.6uH
12V	309K	37.4K	10uH
5.5V	130K	17.14K	3.6uH
6V	143K	18.7K	3.6uH

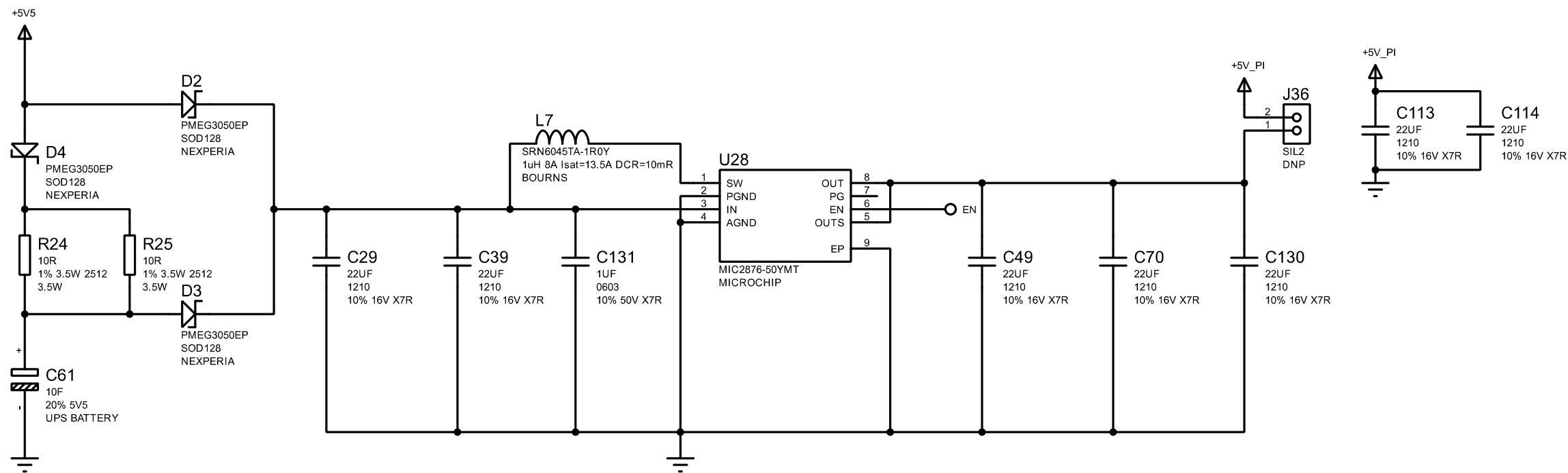
TITLE: D24 Digital		5V
FILENAME: D24 Digital PCBA rev C.pdsprj		
AUTHOR: Peter Watts	PAGE:	14/16
DATE: 08/04/2026	REV:	C



RES VALUES FOR OUTPUT VOLTAGES
INPUT VOLTAGE 3.8V - 40V

OUTPUT	R1	R5	L
1.2V	11K	3.74K	1.5uH
1.5V	19.6K	4.75K	2.2uH
1.8V	27.4K	5.62K	2.2uH
2.5V	47.5K	7.87K	3.3uH
3.3V	69.8K	10.5K	3.3uH
5.0V	115.8K	15.8K	3.6uH
12V	309K	37.4K	10uH
5.5V	130K	17.14K	3.6uH
6V	143K	18.7K	3.6uH

TITLE: D24 Digital	5V5
FILENAME: D24 Digital PCBA rev C.pdsprj	
AUTHOR: Peter Watts	PAGE: 15/16
DATE: 08/04/2026	REV: C



TITLE: D24 Digital		5V_PI	
FILENAME:		D24 Digital PCBA rev C.pdsprj	
AUTHOR:		Peter Watts	PAGE: 16/16
DATE:		08/04/2026	REV: C